

External signals × workforce retention

Are Asteria's retention objectives on track, and do labour-market conditions help explain them? A small data product with honest answers, including the limits of the data.

Bishoy Makhless

github.com/Bishoy-Makhless/Agentic-Engineering-Assessment

Asteria Consumer Products is fictional; the workforce data is synthetic.

3

retention objectives

6

European countries

4

official market signals

2,400

employees, hires 2021–2025

An open question, narrowed to one clear decision

The decision behind the product

Is each retention objective on track, and do market conditions add useful context, without claiming more than the data supports?

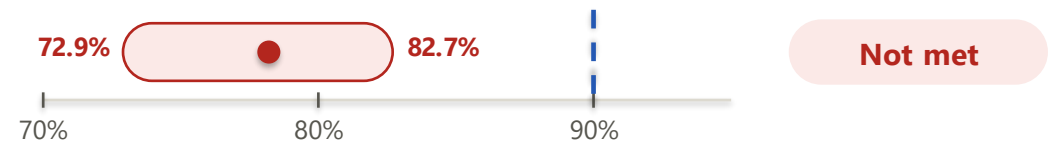
What the product delivers

- A clear status for each objective, with its uncertainty shown
- Official market signals alongside, using only figures already published at the time
- An honest test of whether the two move together, including when they do not
- Full traceability: sources, freshness and exclusions behind every number

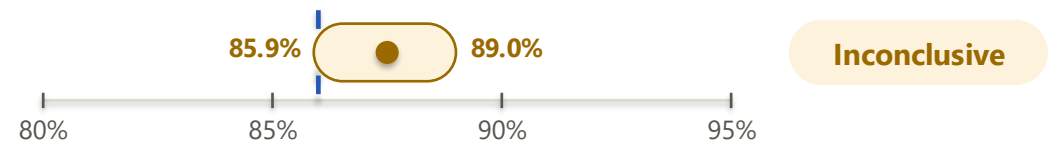
How a verdict is decided: our three results

Each rate has a 95% range. **Met** or **Not met** only when the whole range sits on one side of the target; otherwise **Inconclusive**.

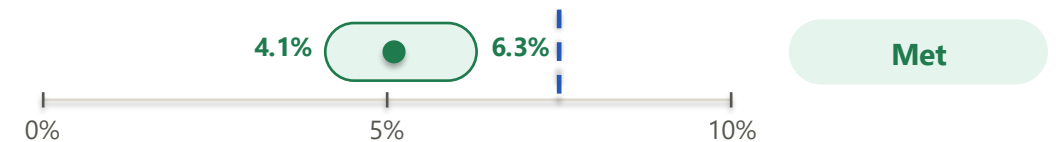
Senior-hire, 12 months · target $\geq 90\%$



New-hire, 6 months · target $\geq 86\%$



Regretted turnover · target $\leq 7.5\%$



Three clear definitions decide every number

New-hire retention

Target: at least 86%

- Retained: still employed six calendar months after the hire date
- Any exit counts, whatever the reason
- Hires too recent to reach six months: **not yet measurable**, never counted as retained

Senior-hire retention

Target: at least 90%

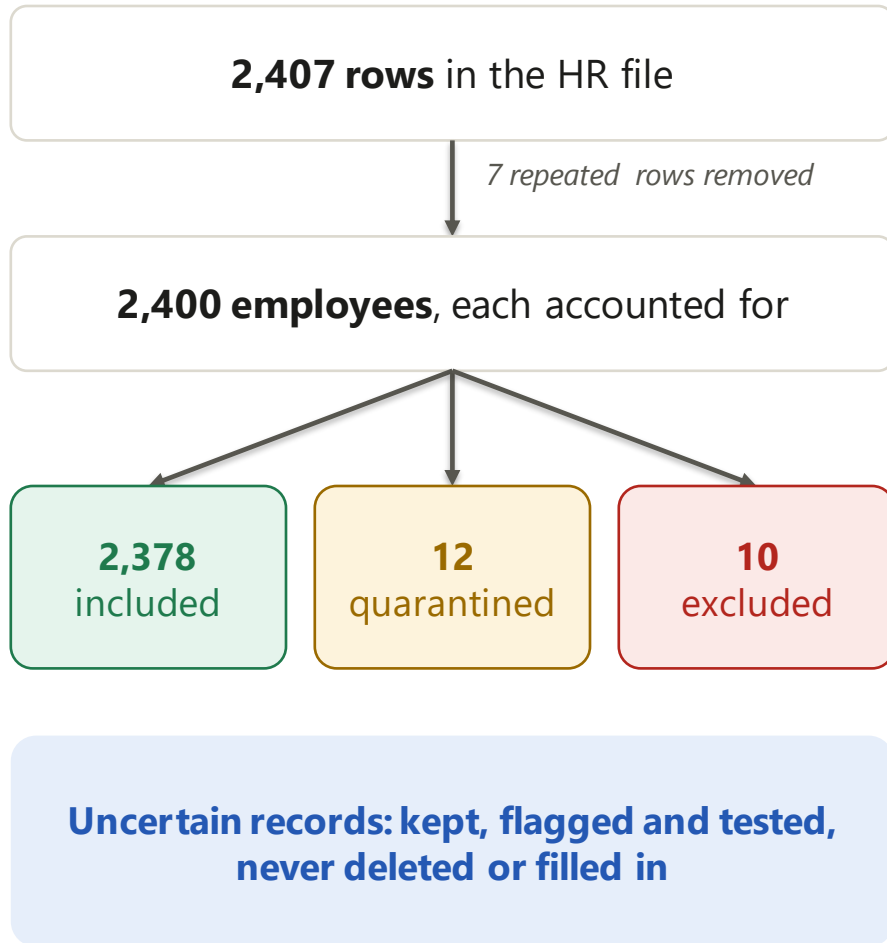
- Same rule over twelve months, applied to Senior Leaders
- 'Senior' left open by the brief, so kept as **one setting**
- With Managers counted as senior: 81.3%, same verdict

Regretted turnover

Target: at most 7.5%

- Confirmed regretted exits over the last twelve months, divided by the average headcount across that year
- Verdicts on December values only, since neighbouring months share eleven months of data
- An unknown 'regretted' answer, never read as 'no'

The data is imperfect on purpose, so nothing is guessed

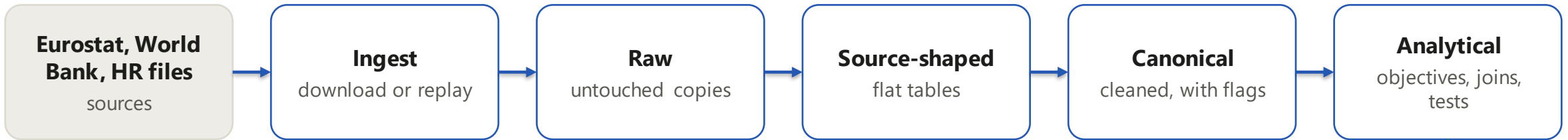


ISSUE IN THE FILE	TREATMENT
Seven employees listed twice	Latest update kept, older row logged
Country codes EL and ROM; nine blanks	Mapped to GR and RO; blanks in company totals only
'Sr Mgmt' as a career level	Treated as Senior Leader
No hire date, or exit before hire (10)	Excluded, each one listed
12 exits exactly 90 days after hire, no type	Quarantined, then tested both ways
2 exits with no 'regretted' answer	Kept as unknown; worst case tested

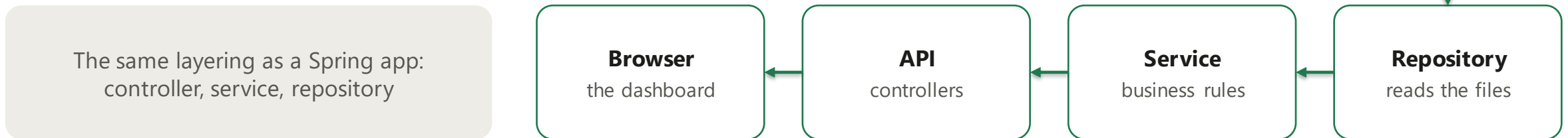
Result: **no verdict changes** when the uncertain records are treated differently

Two flows: one builds the data, one serves it

retention run: the batch job that builds the data

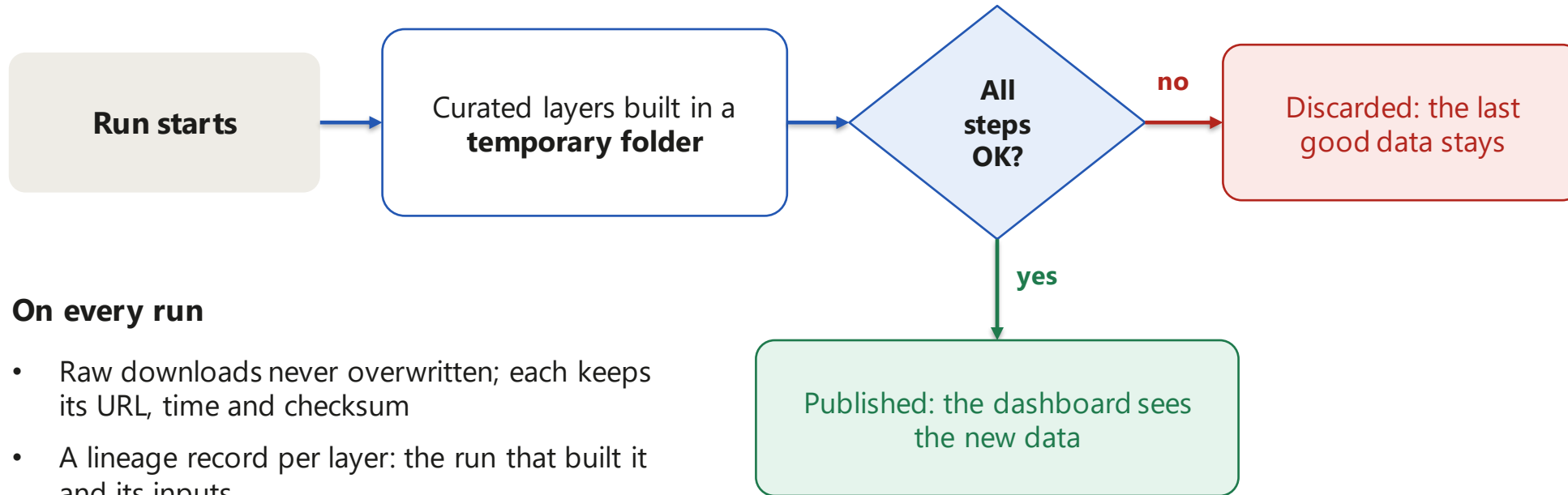


retention serve: the web app that shows it



Python 3.11, pandas and DuckDB SQL on Parquet files, a FastAPI service, a plain HTML dashboard with Chart.js, tested with pytest and Playwright. No database server, no cloud account.

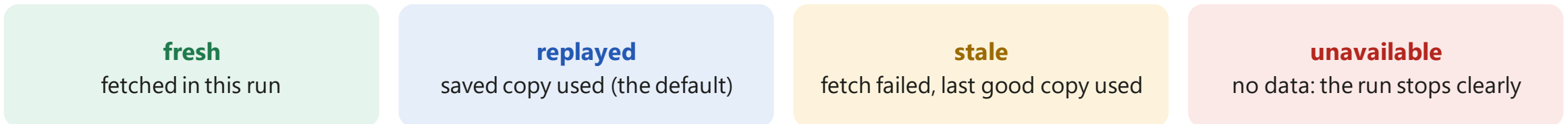
A run is all-or-nothing, and failures are visible



On every run

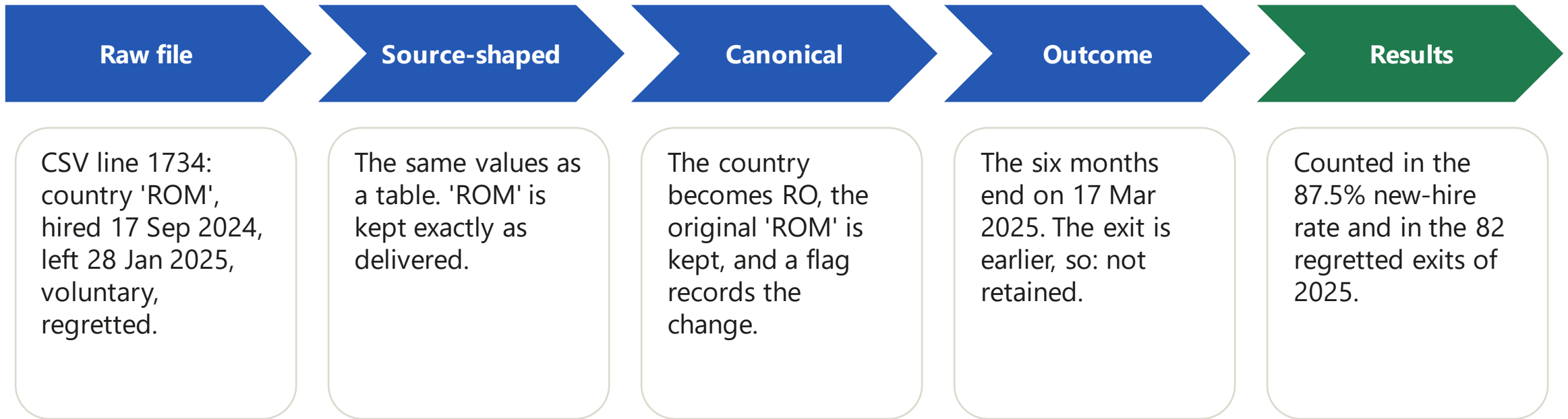
- Raw downloads never overwritten; each keeps its URL, time and checksum
- A lineage record per layer: the run that built it and its inputs

A status for every source, shown on the dashboard:



Every number can be traced back to a line in a source file

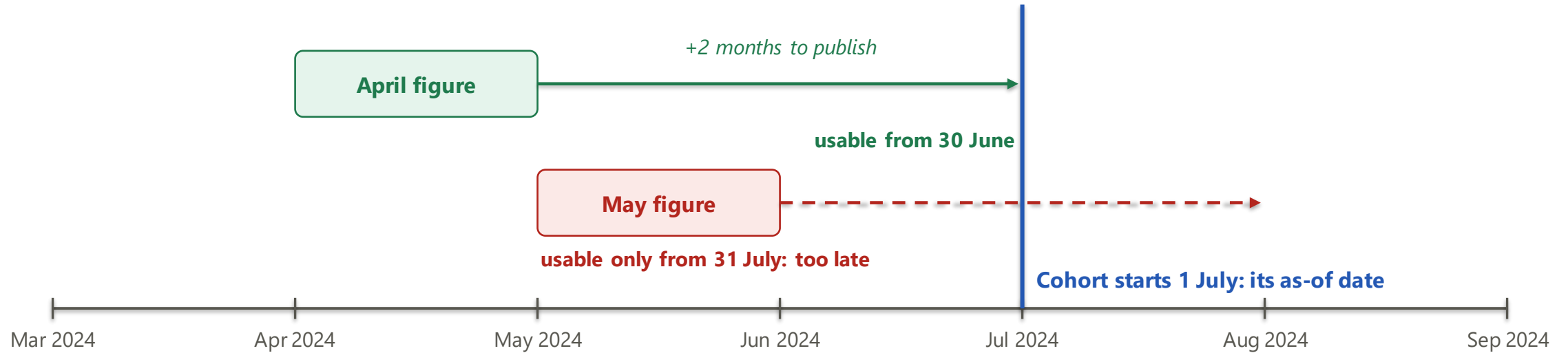
One employee, ACP001733, followed through the layers:



Nothing overwritten on the way: each layer adds to the record, so any value on the dashboard leads back to its source row.

Each cohort only sees data that was already published

Official figures appear weeks or months after the period they describe. Example: the Romanian hires of July–September 2024.



- Usable from: the end of the period plus a fixed lag of **monthly +2, quarterly +3, yearly +7 months**
- Lags measured from the APIs and rounded up: no figure used before its publication

Mixed frequencies stay honest

Yearly GDP for this cohort is the 2022 figure, and it is shown as '**2022, yearly, 19 months old**'. A yearly value is never presented as a new monthly or quarterly one.

186 automated tests guard the riskiest rules

105

unit tests

The cleaning rules, the metric SQL on small hand-made tables, the intervals and verdicts, and the no-future-data join.

41

API tests

Every endpoint, on data built by the real pipeline, including the error responses.

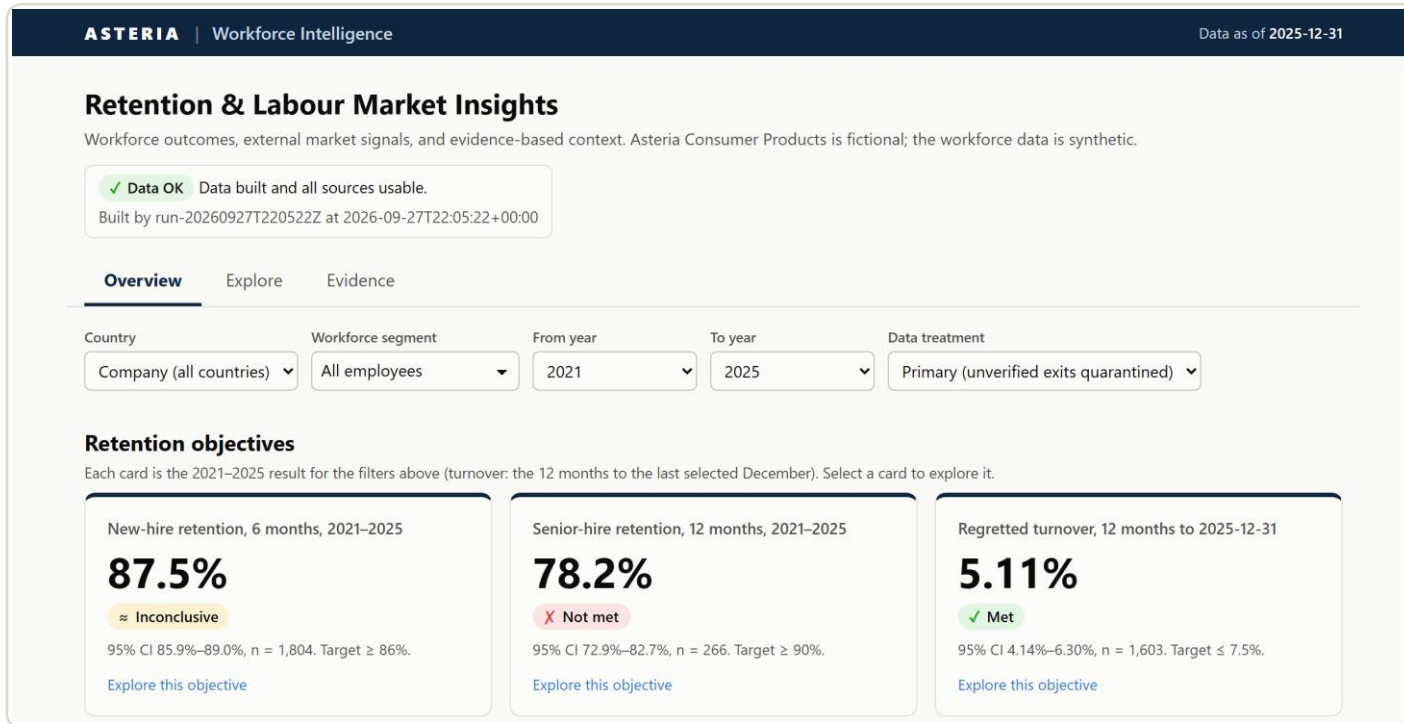
40

browser tests

The real dashboard in Chromium: filters, empty and error states, keyboard use, and a table behind every chart.

- Deliberate breakages of the dashboard, caught by the tests
- 198 dashboard states checked automatically before release; a regression test for every bug found
- The full suite: one command, about a minute and a half

The dashboard answers the brief's four questions



Explore

Country, years, objective and workforce segment filters

Understand

Each objective's status next to the market signals

Challenge

Relationship tests with sample sizes and caveats

Trust

Sources, freshness, exclusions and method

Every number shown with its range, sample size and verdict

Senior retention is below target; the other two need care

Senior-hire, 12 months

78.2%

Not met

208 of 266 senior hires stayed a year. The whole range, 72.9–82.7%, is below the 90% target, in every year and every segment.

New-hire, 6 months

87.5%

Inconclusive

1,579 of 1,804 hires stayed six months. The range, 85.9–89.0%, still includes the 86% target, so the data cannot decide yet.

Regretted turnover

5.1%

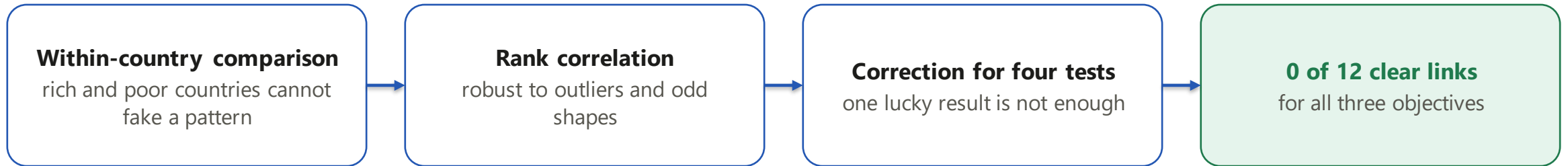
Met

82 regretted exits against an average of 1,603 people in 2025. Below 7.5%, but up from 3.3% in 2024.

Non-finding

None of the 12 formal tests shows a clear link between the market signals and the objectives after correcting for multiple tests. This is a result too, and it is what this synthetic data would predict.

Market signals showed no clear link with retention



	Unemployment	Inflation	Job vacancy	GDP growth
New-hire (primary)	ρ -0.10 Holm p 0.87	ρ -0.12 Holm p 0.87	ρ -0.02 Holm p 1.00	ρ 0.00 Holm p 1.00
Senior-hire	ρ -0.07 Holm p 1.00	ρ +0.20 Holm p 1.00	ρ -0.12 Holm p 1.00	ρ +0.27 Holm p 1.00
Regretted turnover	ρ -0.08 Holm p 1.00	ρ -0.40 Holm p 0.12	ρ +0.02 Holm p 1.00	ρ -0.27 Holm p 0.59

- Inflation and turnover (highlighted): linked on their own ($p = 0.03$), not after correcting for four tests (0.12)
- A steady fall in unemployment over time: time itself could explain a pattern
- Ranges marked exploratory, since the same country appears many times
- Associations only, never causes

A clear link would need Holm p below 0.05. None of the 12 reaches it.

What I chose, what I left out, and why

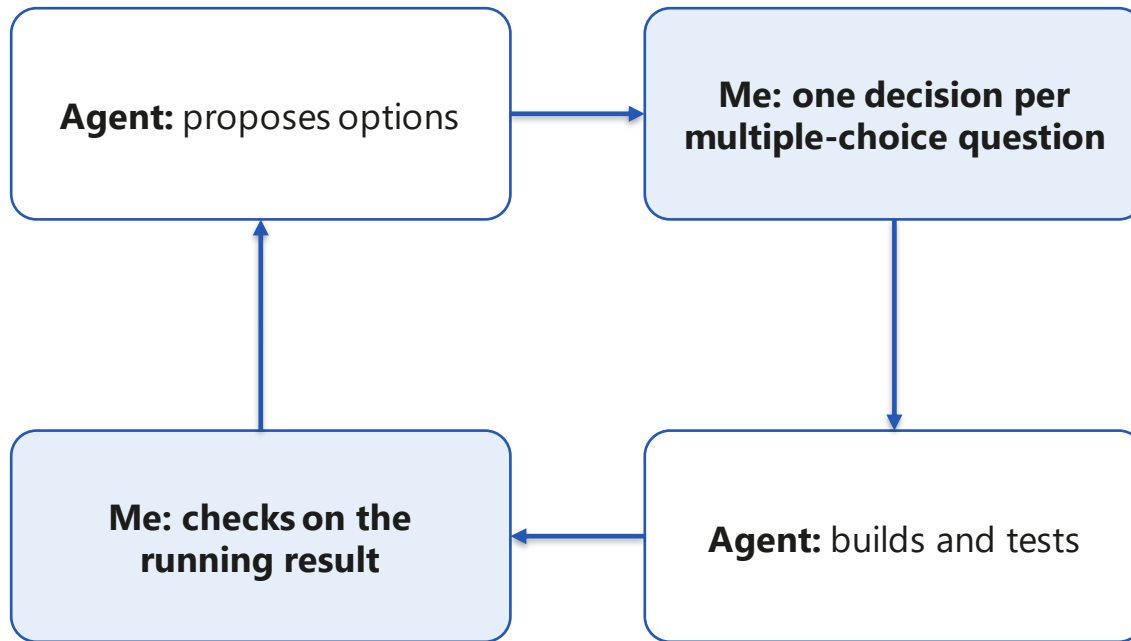
Chosen, and why

- A plain HTML page with a small API instead of Streamlit, so the service can be tested on its own.
- SQL in DuckDB on Parquet files: real SQL, with no database server to install.
- Market data as it was known at hire time, not during the months after, so there is no hindsight.
- Simple, robust statistics instead of regression models, because the samples are small.

Left out, and known limits

- No CI pipeline within the timebox; the tests run locally with one command.
- The HR file seems to hold only people hired since 2020, so early 2021 turnover is inflated.
- The APIs return today's revised figures, not the ones published at the time.
- Each country-quarter has only 7 to 29 hires, so the ranges are wide.

I directed the agent, and every decision stayed mine



Claude Code for the code; ChatGPT for explanations only, with no data shared. Every decision logged with its options and reason.

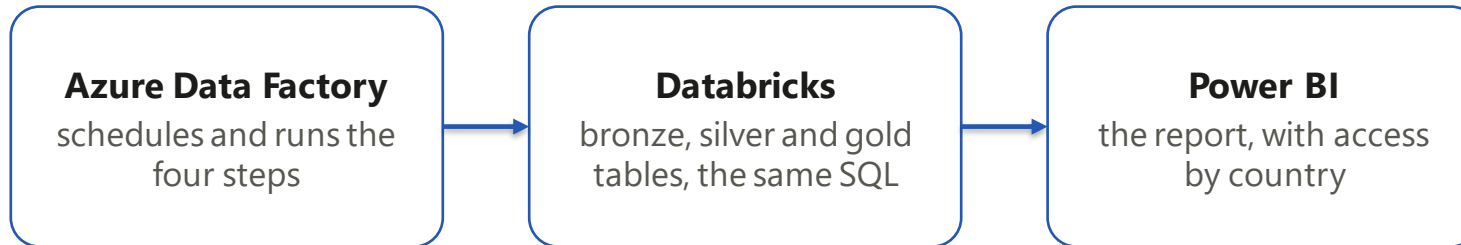
Suggestions I rejected or changed

- Streamlit, replaced by a plain HTML dashboard
- 90-day exits as fake, replaced by quarantine and a test
- Unknown 'regretted' as 'no', replaced by keeping it unknown
- A flashy redesign, rolled back after a look at it running

Mistakes caught by checking

- A guessed dataset code with no data behind it; the real code taken from the provider's own list
- A wrong preview statistic (8.3% instead of 6.6%); the finding withdrawn

From laptop to production, and what would settle more



Key Vault for secrets · alerts on failed or stale runs · Git and CI/CD from dev to test to prod

Same layers, rules and SQL; only the platform changes. DuckDB becomes Spark SQL, the Parquet folders become Delta tables, and the dashboard becomes a Power BI report.

Evidence that would settle more

- HR confirmation of the open assumptions, such as who counts as senior
- Exit reasons from interviews or surveys, to explain why people leave
- About 1,980 mature hires at the same rate, enough to settle the new-hire verdict
- A longer history, and a model that controls for time

SUMMARY

What leaders can take away

Senior-hire retention

78.2%

Not met

Clearly below the 90% target, in every year and every segment.

New-hire retention

87.5%

Inconclusive

Close to the 86% target; about 180 more mature hires would settle it.

Regretted turnover

5.1%

Met, but rising

Within the 7.5% limit, up from 3.3% in 2024; worth watching.

Market signals: no clear link with any objective (0 of 12 tests). Useful as context, not as an explanation.

Next: confirm the open assumptions with HR, add exit reasons, and extend the history.

Thank you.